

# B.E. 4th Semester Theoretical Examination, 2022-2023

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Subject: Discrete Mathematics | Course: PCC-IT 401

## Q1. Propositional Logic

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- **Sentence:** Declarative statement strictly True or False.
- **Conjunction ( $p \wedge q$ ):** "AND" operator. True only if both are true.

## Q2. Equivalences and Tautologies

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- **Tautology:** Compound proposition always True.
- $p \vee \sim p$ : Law of Excluded Middle (Always True).

## Q3. Set Theory and Functions

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- **Equivalence Relation:** Reflexive ( $aRa$ ), Symmetric (if  $aRb$  then  $bRa$ ), Transitive.

## Q4. Posets and Lattices

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- **Poset:** Reflexive, Antisymmetric, Transitive relation.
- **Lattice vs Poset:** A Lattice requires every pair to have a unique LUB and GLB. Not all posets have this, so not all are lattices.

## Q5. Group Theory

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- **Group is Abelian** iff  $(a \circ b)^{-1} = a^{-1} \circ b^{-1}$ : Normal inverse rule is  $b^{-1} \circ a^{-1}$ . If abelian, this flips.
- **4th Roots of Unity:**  $\{1, -1, i, -i\}$  forms a commutative group under multiplication.

## Q7. Graph Theory

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- **Handshaking Theorem:**  $\sum \text{deg}(v) = 2|E|$ .
- **Isomorphism:** Bijective mapping preserving adjacency between two graphs.

## Q8. Graph Sub-types

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- **Regular Graph:** Every vertex has same degree.
- **Bipartite Graph:** Vertices split into two disjoint sets; edges only cross between sets.
- **Chromatic Number of Tree:** 2.